

National University of Computer and Emerging Sciences, Lahore Campus			
	Course:	Machine Learning for Robotics	Course Code: CS 5096
	Program:	BS (Computer Science)	Semester: Spring 2026
	Date:	09-02-2026	Total Marks: 15
	Section:	B	Weight:
	Exam:	Quiz 1	Roll No:

Use one iteration of gradient descent to update the weights of a multivariable linear regression model.

Initial values: $\theta_0 = 0.5$ $\theta_1 = 0.5$ $\theta_2 = 0.5$ $\alpha = 0.01$

The cost function is given by:

(10)

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Find the updated values of θ_0 and θ_1 and θ_2 after this iteration.

Driving Experience (years) (x_1)	Car Age (x_2)	Monthly Auto Insurance Premium (Y)
5	3	64
2	8	87
12	1	50

complete the table and show equations used

Q2)Normal Equation

(5)

Assume you have m training examples and n features (excluding the bias term).

a) What is the dimension of the design matrix X ?

b) What is the dimension of $X^T X$

c) What is the dimension of the parameter vector θ ?